

Project Proposal.

Secure Guest and Staff Network

For my project, I've decided to build a virtual network for a local library called "Book Club". The network serves the purpose of allowing staff to perform duties such as managing inventory and checking in and out books, as well as allowing customers to browse the internet on personal devices or provided desktops.

In this project, there will be 3 kinds of networks included in the greater network. A wired network for staff use, a wired network for public PC use, and a wireless network for personal devices.

This design demonstrates how a router, switch, and Access Point work together to provide automated configuration (DHCP), name resolution (DNS), network address translation (NAT), and secure communication (WPA2/WPA3) for both wired and wireless clients in a hybrid local network environment.

To manage this environment, we are using one router, one switch, one wireless Access Point, and four devices/machines. All wired and wireless traffic must flow through the central Gateway Router to reach the External Internet.

1. The Router (Gateway/DHCP): The "Traffic Cop" of the library. It links all internal library devices together, assigns IP addresses to everyone (Staff, Public PCs, and BYOD Guests), and uses NAT so internal devices can securely access the external world.
2. The Switch: A central hub that physically connects the wired Staff PC, the wired Public PCs, the Catalog Server, and the Wireless Access Point.
3. WLAN Access Point (AP): This device connects to the main switch and creates the wireless network ("BookNet_Guest"). Its primary job is to translate wired network data into radio signals for smartphones, laptops, and tablets.

Internal Devices:

1. Staff Desktop (Wired): Front desk computer for administrative work.
2. Public Desktop (Wired): Desktop computer in the lobby for visitors without personal devices.
3. Library Data Server (Wired): Single server hosting the Catalog (HTTP/HTTPS) and directory information (DNS).
4. Community BYOD Devices (Wireless): A generic laptop representing multiple users connecting to the wireless network using WPA2/WPA3 security.